Lead71 Final Validation Package

CardLens AI mapping and capture safeguards | 30 October 2026

Purpose and decision

This package records the available code-level and automated synthetic evidence for the staging implementation. It contains no credentials, configuration screenshots, business-card images, or client data. The project team requested that AI mapping remain out of Lead71 production until this package is reviewed.

Recommendation: NO-GO for production deployment at this time. The staging source has automated test coverage, but the deployed staging OCR route returned HTTP 500 during the latest check. A live hosted end-to-end card review and save has therefore not been established by this evidence package.

Staging reference

Branch / commit	staging / c2a4801 - Add AI mapping rollback and fallback cap
Staging URL	https://cardsnap-vision71-staging.vercel.app
Deployment checks	Vercel checks for the staging commit were reported successful. This indicates build/deployment checks only; it does not establish that the OCR route works.
Latest hosted check	GET /api/health returned HTTP 200. POST /api/ocr returned HTTP 500 (FUNCTION_INVOCATION_FAILED). Request ID: sin1::4v277-1793360796518-eb044e31fc63.

Resource separation

Per project team confirmation: Haroon Bhai owns the staging MongoDB, which is separate and isolated from the main production database. Hassan owns the staging Google Sheet; Ali Bhai owns the main production Google Sheet. These ownership and separation statements are recorded without displaying identifiers or secret values.

The local configuration audit indicated staging-oriented URL/database settings and a staging email allowlist. It did not independently compare the deployed Vercel environment, MongoDB identifiers, Sheet identifiers, email sender credentials, or all configuration values. No live staging Sheet write was performed in the checks summarized here.

Synthetic test evidence

The staging commit automated suite passed 22 of 22 tests. The scenarios below used synthetic inputs and injected in-memory/test service adapters. Reported milliseconds are measured Node test-case runtimes, not production provider latency. No real Gemini/OCR provider call, external database write, Google Sheet write, or client data was used.

Scenario	Observed result	Measured test time	Review / save result
AI mapping enabled	AI classification success path returned AI mode using the test classifier.	~0.6 ms	Success-path mapping covered. This automated case did not submit a live record.
AI mapping disabled	ENABLE_AI_MAPPING=false bypassed classifier; manual parser returned HTTP 200. Classifier call count: zero.	4.6 ms	Manual route remained usable in the automated handler test.
AI API failure	Synthetic 503 led to manual parser result.	1.2 ms	Manual result available for review in test response.
AI timeout	Synthetic timeout led to manual parser result.	0.9 ms	Manual result available for review in test response.
Invalid AI response	Invalid synthetic AI payload led to manual parser result.	1.0 ms	Manual result available for review in test response.
Quota / rate limit	Synthetic 429 led to manual parser result.	0.8 ms	Manual result available for review in test response.
No credit / rejected request	Synthetic 402 led to manual parser result.	0.9 ms	Manual result available for review in test response.
AI monthly cap reached	AI mapping returned manual result; separate manual request passed.	1.1 ms	Manual capture path was not blocked by the AI cap.
Concurrent cap	Cap set to one in test: one AI test call and one manual path; both returned HTTP 200.	14.4 ms	Both requests completed in test.
Synthetic review and save	Synthetic contact parsed as Synthetic Name / synthetic@example.test. Test submitDirect returned submitted.	26.9 ms	Saved to in-memory Sheet fixture only; no real Sheet write.

Manual/OCR availability when AI limits are reached

The tested AI monthly cap (AI_MAPPING_MONTHLY_CAP; default 900 in the staging source) gates AI mapping requests. With the AI cap reached, the tested path returned manual parsing and a manual request still succeeded. Turning AI mapping off also bypasses the classifier. Existing general OCR abuse-rate guards are separate from the AI mapping cap and can still limit OCR requests when their own hourly thresholds are reached; therefore the evidence supports isolation from the AI cap, not unlimited OCR availability under every limit.

Security and evidence boundaries

Production Gemini key

No production Gemini key was used in the synthetic tests, added by this work to source code, or exposed in this package. A repository history scan found no Google API-key-shaped literal in the inspected source/history. The Vercel production environment settings and key vault were not accessible for an independent check; this package therefore cannot certify that a production key is absent from Vercel or confirm its current secure storage/usage status.

Environment separation

The staging database and Sheet ownership/separation statements above are provided by the project team. No secret values or environment screenshots are included. The automated tests used local/in-memory fixtures. Hosted Vercel configuration separation for database, Sheet, email, and other configuration values was not independently inspected.

Evidence distinction

Implemented and covered in automated staging-source tests: AI mapping flag bypass, manual fallback for tested AI failure classes, AI cap isolation, concurrency cap behavior, and synthetic handler-to-save behavior.

Hosted staging status: the commit is identified above and deployment checks were reported green, but the latest hosted OCR endpoint returned HTTP 500. The automated synthetic suite does not substitute for a successful hosted staging scan, review, and save.

Production boundary

No production deployment, production API-key use, or Aventure/client card data use occurred as part of this validation. Keep AI mapping out of production until the staging OCR error is resolved and a synthetic live staging scan can be reviewed and saved end to end.

Prepared for review

This is the clean replacement validation package. It contains no screenshots, credentials, client records, or fabricated card imagery. All synthetic test data was generated within the automated test suite.